Ctrl+Z Burnout

Academic Decision Simulator

Custom Track Proposal & Target Principle:

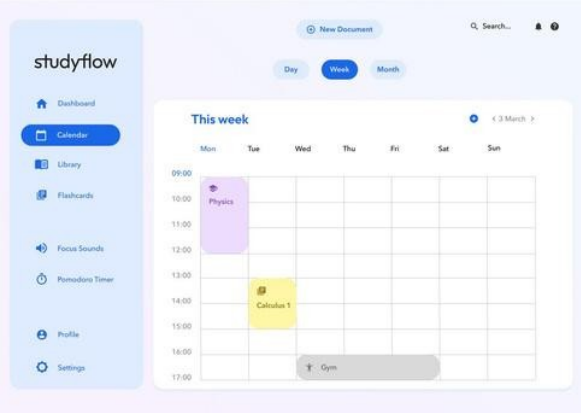
- **The Idea:** A web-based simulator that merges academic course advising (prerequisites) with dynamic weekly study-workload prediction to prevent student burnout. Target audience: University students.
- **Target HAI Principle:** "User Control and Agency". The system provides a live constraints dashboard where students can accept, reject, or undo any AI suggestion, ensuring they always have the final say.

Milestone 1:

Concept Drafts - The Approach:

Below are the full raw outputs from two different AI tools generated using the same prompt, presented side-by-side for direct comparison of their interaction flows.

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Academic Decision Simulator UX Research & AI Interaction Design Concept · Team Ctrl+Z Burnout The Academic Decision Simulator is an AI chatbot that helps university students make informed course-selection decisions while simulating the study workload those decisions create. It combines academic prerequisite checking, personalized schedule generation, and workload-aware planning to help students make decisions without losing control over their academic lives. Core design principle: User Control and Agency. The AI recommends, explains, and simulates. The student decides, edits, accepts, rejects, and reverses decisions at every critical stage. **1. Concept approach: Conversational planner + interactive dashboard**	**Academic Decision Simulator: Core Interaction Flow Concept** Team Ctrl+Z Burnout | HAI Principle: User Control and Agency **1. Concept Approach** Style: Constraint-first hybrid ("Conversational Dashboard") The interaction pairs a short, guided conversation with a persistent, live Workload Dashboard. The chat handles intent and explanation. The dashboard handles numbers and control. Why this fits: ● Constraints come before suggestions. Students set their limits first, so the AI works inside their boundaries instead of proposing something they then have to push back against.

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Use a hybrid interaction model: a conversational chatbot for guidance, supported by a persistent constraint dashboard and a visual schedule simulator.  Conversational guidance The chatbot asks questions one at a time, explains prerequisite requirements in simple language, and helps students understand trade-offs instead of making decisions for them.  	● Every AI output is a proposal, never a decision. The AI's language is "Here's an option," not "Here's your schedule." Nothing is committed until the student says so. ● Simulation is visible and manipulable. The burnout forecast (weekly hours, peak weeks, recovery gaps) is a live meter the student can poke, not a hidden score. ● Prerequisites are advisory, not gatekeeping. The AI flags a missing prerequisite, explains it, and offers alternatives, but the student can override with a clear warning (e.g., "I have advisor approval"). The tone is a collaborative co-pilot: it offers, explains, and defers. **2. Step-by-Step Flow** **Step 1: Onboarding (≈60 seconds, skippable)** Goal: Establish who is in charge and gather minimum context. 1. Welcome message with a control statement: *"I'll suggest course plans and simulate your workload. You decide everything. You can reject, edit, or undo anything I propose."* 2. Quick context capture (tappable chips, no long forms): a. Program/major, current semester, completed courses (import transcript or tick a checklist)

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Constraint dashboard Students can always see and edit their available study hours, class timetable, work commitments, deadlines, and workload preferences.  Dynamic simulation Every accepted change or edited constraint triggers an updated workload simulation. Students compare alternatives before committing to a final plan. Why this approach? A purely conversational interface may hide important constraints, while a dashboard-only interface may feel complicated. Combining both makes the system understandable, transparent, and flexible. **2. Step-by-step interaction flow** **Step 1 — Onboarding** Understand the student's goals, academic context, and preferred way of planning. **Step 2 — Set constraints** Student defines the limits the AI must	b. Optional: "Anything I should know?" (job, commute, caregiving, health) 3. Transparency note: what data is used and a "Delete my data" link. 4. Skip option: "Skip for now" is always visible. The bot continues with defaults and labels them as assumptions. **Step 2: Setting Constraints (the "Guardrails" panel)** Goal: The student defines their limits before any suggestion appears. Key design details: ● Each constraint is labeled Hard (AI may never violate) or Soft (AI may exceed only with explicit warning). The student toggles this per constraint. ● The AI reflects constraints back in plain language: *"So I'll keep you under 30 hours/week, with Sundays protected. Right?"* with [Yes] [Edit] buttons. **Step 3: AI Generation (Propose, don't impose)** Goal: Present options with visible reasoning and a workload simulation. 1. The AI generates 2-3 distinct plan options, not one "best" answer, for example: a. Plan A: Balanced (14 credits, peak week 28 h) b. Plan B: Fast track (16 credits, peak week 36 h, flagged as risky) c. Plan C: Light (12 credits, peak week 21 h, delays graduation by 1 term)

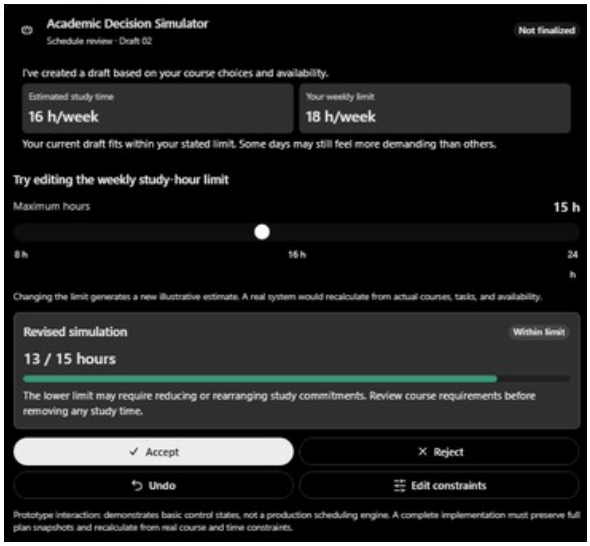
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respect. ● Maximum weekly study hours ● Available days and time slots ● Courses under consideration ● Work, commuting, and personal commitments ● Preferred workload intensity Agency checkpoint: The student can edit these values at any time. **Step 3 — Validate course eligibility** The AI checks course prerequisites against the student's academic record and the official course information available to it. ● Eligible courses ● Missing prerequisites ● Unverified requirements ● Potential scheduling conflicts Agency checkpoint: The student reviews warnings and can remove or replace course choices. **Step 4 — Generate schedule options** The AI produces alternatives, not a single mandatory answer. ● Option A: Balanced workload ● Option B: More flexible study blocks ● Option C: Higher course load, if feasible Agency checkpoint: The student chooses which options to compare. **Step 5 — Simulate and review** Display weekly hours, daily workload, deadlines, free time, and potential overload	2. Each plan card shows: a. Course list with prerequisite status (met / missing / override needed) b. Weekly workload heatmap (weeks 1-15, color-coded against the student's own limit) c. Burnout risk indicator with a "Why?" expander (e.g., "Weeks 7-9 stack 2 midterms + 1 project") d. Trade-offs in one line (e.g., "Saves a semester, costs 6 more hours/week") 3. What-if sandbox: the student drags a course in or out and the heatmap updates instantly, labeled "Simulation only, not saved." 4. The AI states its confidence and assumptions (e.g., "Estimated hours are based on past student averages for this course"). **Step 4: Refinement Loop** Goal: Iterate until the plan feels right. ● The student reacts through buttons or natural language ("Make week 8 lighter"). ● The AI responds with a diff, showing what changed and why, rather than silently regenerating. ● Each iteration is saved to a visible Version Timeline (v1, v2, v3...). **Step 5: Finalization** Goal: An explicit, informed commit. 1. Review screen: final course list, workload summary, any overrides or warnings the student accepted.

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warnings. Agency checkpoint: Accept, Reject, Undo, edit constraints, or request a different simulation. **Step 6 — Finalize the plan** The student reviews the chosen courses and schedule, confirms the plan, and exports it or saves it for later. Agency checkpoint: Final confirmation is always explicit. No course registration or external commitment happens automatically. **3. Agency mechanisms: Accept, Reject, Undo, and Edit** These controls should be available during schedule review and, where relevant, earlier course-selection steps. Each control must produce a predictable result. **Accept — Keep this change** Commits selected change to the working plan When: After the AI proposes a schedule, course swap, or time adjustment. Behavior: ● Accept a whole schedule or one specific change. ● Update the working plan and recalculate weekly and daily workload. ● Keep the previous version available for Undo. ● Do not treat acceptance as final registration or irreversible commitment. Example: “Accept moving Database Systems study time to Wednesday.” The AI	2. Explicit confirmation: *"Save this as your plan?"* [Save plan] [Keep editing] 3. Export options: PDF, calendar sync, "share with advisor" (the AI never contacts an advisor automatically). 4. Post-save reassurance: *"Saved as draft. You can reopen and change it any time."* Finalization is a soft commit, not a lock. **3. Agency Mechanisms** **3.1 Accept: Yes, take this, but I'll still shape it** Principle: Acceptance can be partial and granular. The student can accept 4 of 5 courses without accepting the whole plan. **3.2 Reject: "No, and here's why (optional)"** Rules: ● Giving a reason is optional. One tap to reject, no justification required. ● The AI never argues or repeats a rejected suggestion within the session unless the student asks. ● Rejected items go to a "Dismissed" tray where they can be restored. **3.3 Undo: "Ctrl+Z for everything" (brand-aligned)** Safeguards: ● Undo is available on every state change, including constraint changes, not only schedule changes. ● A brief toast appears after each action: *"Course removed. [Undo]"* (visible for ~8 seconds, with the full history still

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updates the draft and recalculates the Wednesday workload. **Reject — Keep my current plan** Dismisses the proposed change When: The AI suggests an unsuitable schedule or a course option the student does not want. Behavior: ● Preserve the current working plan. ● Ask an optional follow-up: “What would you prefer to change?” ● Offer another suggestion based on the student's feedback. ● Never repeatedly reintroduce a rejected option without a meaningful reason. Example: “Reject Option B — I need my evenings free.” The AI keeps the existing plan and searches for alternatives that respect evening availability. **Undo — Restore the previous version** Reverses the last change When: Immediately after accepting a change, editing a constraint, or applying a new schedule. Behavior: ● Restore the previous working plan and associated constraint values when appropriate. ● Recalculate workload indicators. ● Display a confirmation message identifying what was reversed. ● Keep a version history so the student	available afterward). ● Destructive actions (e.g., "Delete all versions") require confirmation. Everything else is reversible without it. **3.4 Edit Constraints: "Change the rules, watch the plan respond"** When: at any point, from a persistent Guardrails panel (sidebar on desktop, bottom sheet on mobile). Constraints are never locked behind onboarding. How it works (live recalculation loop): 1. Student adjusts a constraint, e.g., drags max weekly hours from 30 → 24. 2. The dashboard shows an instant preview of impact before applying: *"At 24 h/week, Plan A exceeds your limit in weeks 7-9. 2 options: drop CS301 or swap to the lighter section."* 1. Student chooses [Apply & regenerate] or [Cancel]. 2. The AI returns a diff view (what changed and why), not a full replacement. 3. The change is added to the Version Timeline and is undoable. Additional controls: ● Hard/Soft toggle per constraint (see Step 2). ● Locked items persist: anything the student accepted/locked stays put unless the student unlocks it. If a constraint change would force a conflict with a locked item, the AI asks rather than

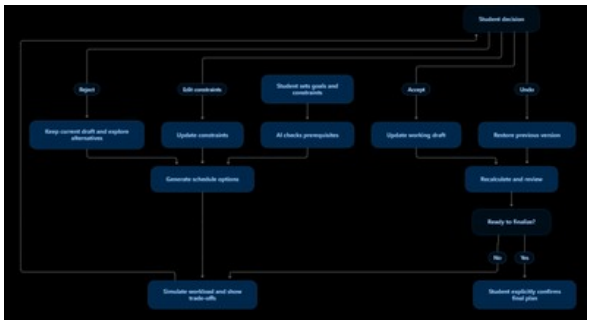
ChatGPT	Claude
can return to an earlier draft. Example: “Undo: Wednesday's study block was moved back to Tuesday.” The earlier schedule reappears without requiring the student to rebuild it manually. **Edit constraints — Change the rules** Re-simulates the plan using new limits When: At any stage, especially when reviewing a schedule that feels too demanding or no longer fits the student's commitments. Behavior: ● Allow direct editing of weekly study-hour limits, available time slots, course choices, and other constraints. ● Show which existing schedule elements may be affected before applying the changes. ● Re-run the simulation and explain what changed. ● Preserve the prior version so the student can Undo or compare plans. Example: “Change my maximum study time from 18 to 14 hours a week.” The AI generates a revised plan, flags any courses or workload requirements that no longer fit, and lets the student decide what to adjust next. **Proposed interaction mockup** The following is an example of how the student could interact with the simulator. The hours and courses are illustrative, not actual academic recommendations.	overriding: *"This conflicts with your locked course. Unlock it, or relax the limit?"* ● Natural-language edits work too ("I got a part-time job, cap me at 20 hours") and the AI converts them into a visible constraint change the student confirms. **4. Agency Safeguards (Cross-Cutting)** ● Explain on demand: Every suggestion has a "Why this?" link showing the reasoning and data source. ● No dark patterns: no urgency language ("Register now!"), no default-selected risky options, and no guilt framing around burnout. ● Override with accountability: the student can override prerequisite or workload warnings, and the override is logged transparently, never blocked. ● AI humility: workload estimates display a range (e.g., "24-30 h") rather than false precision, with a "These are estimates; adjust if you know better" prompt. The student can enter their own hour estimate for any course and the simulation uses it. ● Advisor handoff: the AI states that it is not an official advisor. Prerequisite checks should be verified with the university. **5. Quick Reference: The Flow at a Glance** Onboard → Set Guardrails → AI proposes 2-3 plans → Student

ChatGPT



4. How changes dynamically affect the AI

The system should use a human-in-the-loop feedback cycle, where student actions are inputs to the next simulation rather than mere reactions to a static recommendation.



Example of a dynamic change

1. Initial state: The student has selected five courses and set a maximum of 18 study hours per week.
2. AI recommendation: The simulator estimates 16 hours and displays the distribution across the week.

Claude

reviews

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└──── Edit constraints / Reject / Lock / What-if

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Accept (partial or full)

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Review summary + warnings

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Save draft (reopenable)

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↶ Undo / Version Timeline available at every step

6. Suggested Validation Questions (for your usability testing)

1. Can participants find and use Undo without prompting? (Target: >90%)
2. Do participants feel the plan is "theirs" after finalizing? (Post-task Likert, 1-5)
3. When rejecting a suggestion, do participants feel pressured to justify it?
4. Do participants understand why a week was flagged as high-risk?
5. Do they trust the workload estimates, and do they use the "enter my own estimate" option?

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3. Student intervention: The student changes the limit to 14 hours and rejects an evening study block. 4. AI response: The simulator rebuilds the schedule around the new limit and availability. If the courses cannot reasonably fit, it explains the conflict and presents alternatives. 5. Student decision: The student accepts one alternative, undoes the change if needed, or edits the constraints again. 6. Finalization: The student explicitly confirms the chosen draft after reviewing prerequisite status and workload warnings. An important rule: the AI must never silently relax a student's stated constraints to make a schedule appear feasible. If requirements conflict, it should explain the conflict and ask which trade-off, if any, the student wants to make. **5. UX rules that make User Control and Agency real** **6. Success criteria for UX evaluation** Test whether students can use the system to make informed decisions independently, rather than simply whether they like the chatbot. ● Constraint control: Can students locate and change their weekly hour limit without assistance? ● Reversibility: Can they undo an accepted change and correctly identify the	

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restored plan? ● Comprehension: Can they explain why a suggested course is eligible or why a schedule exceeds their limits? ● Decision confidence: Do they understand the alternatives and feel free to reject the AI's recommendation? ● Transparency: Can they distinguish verified prerequisites from missing or uncertain information? ● Workload awareness: Can they identify scheduling pressure points before finalizing their plan? **Final design recommendation** For Ctrl+Z Burnout, make the product feel less like an AI that tells students what to do and more like a decision simulator that helps students understand the consequences of their own choices. The key interaction loop is: Set constraints → Simulate options → Inspect trade-offs → Accept, Reject, Undo, or Edit → Re-simulate → Confirm. That loop makes User Control and Agency visible in the interface, enforceable in the system behavior, and measurable through usability testing.	

Strengths & Weaknesses:

Concept Draft 1 (ChatGPT): Conversational Dashboard

- **Strengths:**

- Introduces an innovative "Conversational Dashboard" that merges chat with a live, persistent constraints panel.
- Excellent interactive UI mechanics, utilizing sliders and a "Version Timeline" for seamless Undo functionality.
- Provides granular control, allowing students to accept, reject, or lock specific courses without rejecting the entire schedule.

- **Weaknesses:**
  - The continuous real-time recalculation loop might be overly complex to prototype effectively.

Concept Draft 2 (Claude): 6-Step Guided Flow

- **Strengths:**
  - Offers a highly structured, easy-to-follow 6-step user journey.
  - Strong emphasis on academic accuracy, strict prerequisite validation, and identifying scheduling conflicts.
  - Enforces strict AI boundaries (e.g., the AI must never silently relax user constraints to make a schedule fit).
- **Weaknesses:**
  - Lacks detailed focus on visual UI mechanics and visible Undo interaction states compared to Draft 1.

Prompt Log:

#	Tool	Exact prompt	What it gave	What we changed and why
1	ChatGPT	Act as a UX Researcher and AI Interaction Designer. I am building an AI chatbot named "Academic Decision Simulator" (Team: Ctrl+Z Burnout). The target users are university students. The chatbot merges academic course advising (checking	Generated Draft1: A Conversational Dashboard" style with a live constraints panel, slider inputs, and a "Version	**Change:** Extracted only the interactive UI mechanics (like the version timeline and granular accept/reject) to use in our final design. **Why:** To maximize the "User Control and Agency" principle, making it easy for

#	Tool	Exact prompt	What it gave	What we changed and why
		prerequisites) with a dynamic study-workload simulation to prevent student burnout. Mandatory HAI Principle: "User Control and Agency". Please draft a core interaction flow concept for this chatbot. Your concept must clearly show:1. Concept Approach: Briefly describe the style of interaction (e.g., highly conversational, constraint-dashboard first, or guided wizard).2. Step-by-Step Flow: Outline the user journey from Onboarding, to setting constraints (e.g., max weekly study hours), to AI generation, and finalization.3. Agency Mechanisms: Explicitly detail exactly *how* and *when* the user can use "Accept", "Reject", "Undo", and edit their constraints to dynamically change the AI's schedule suggestions.	Timeline" for Undo functionality	students to reverse actions, but we discarded its complex real-time loop to keep prototyping manageable.
2	Claude	Same prompt as above	Generated Draft2: A highly structured 6-step guided flow emphasizing	**Change:** Adopted its strict backend rules regarding AI boundaries but replaced its basic UI steps with ChatGPT’s interactive

#	Tool	Exact prompt	What it gave	What we changed and why
			strict academic validation and strong AI boundaries (AI never silently relaxes constraints).	dashboard. **Why:** Claude's strict rules ensure academic accuracy and true user agency, but merging it with ChatGPT's UI gives the user a better visual experience for the "Undo" and constraint-editing tools.

Idea for Approval (Custom Track Proposal):

We propose the **‘Academic Decision Simulator’**, a custom track that combines Academic Advising with a Study Workload Coach for university students prone to academic burnout. While standard advising tools simply auto-generate schedules based on prerequisites, our idea is unique because it works as a dynamic workload simulator.

Students interact through a Conversational Dashboard, a chat interface paired with a live, persistent constraints panel. The system estimates the workload of each course, factors in exams, assignments, and major projects, and visualizes the result in a Semester Heatmap and a Workload Pressure Indicator that compares expected load with the student’s stated capacity (presented as a range, never as a medical diagnosis).

Students can also explore GPA and graduation what-if scenarios, plan a multi-semester roadmap that highlights bottleneck courses, and simulate dropping a course mid-semester (“Ctrl+Z for the Semester”) to see the impact before taking real action. A weekly 30-second check-in and an actual-vs-planned calibration loop keep the estimates personal and offer opt-in campus counseling resources when needed.

Driven by the **‘User Control and Agency’** HAI principle, the system never finalizes a schedule: students can accept, reject, or lock individual courses and use the Version Timeline to undo any change, retaining full authority over both their academic path and their well-being.

Expanded System Features

Based on the chosen hybrid AI concept, our team expanded the scope into the following 12 core features to distribute the workload across our 5-member team:

1. **Conversational Dashboard:** A chat interface combined with a live, persistent statistical dashboard.
2. **Course Workload Estimation:** Predictive hourly estimates for each enrolled course.
3. **Event-based Inputs:** Tracking system for exams, assignments, and major projects.
4. **Semester Heatmap:** A visual calendar identifying high-pressure (peak) academic weeks.
5. **Workload Pressure Indicator:** Compares expected academic load with the student's stated capacity (presented as a range, explicitly avoiding medical diagnosis).
6. **GPA & Graduation What-if:** Calculates the workload trade-offs required to reach a specific target GPA.
7. **Multi-semester Roadmap:** Plans the full graduation path, highlighting bottleneck courses and their cascading effects if delayed.
8. **Rewind & Drop Simulator ("Ctrl+Z for the Semester"):** Allows a student to simulate dropping a course mid-semester to visualize the impact on their workload and graduation plan before taking actual action.
9. **Calibration (Actual vs Planned):** A feedback loop where the system adjusts future estimates based on the student logging their actual study hours.
10. **Weekly 30-second Check-in:** A brief early-warning prompt regarding energy and stress levels, offering opt-in campus counseling resources if needed.
11. **Granular Accept/Reject:** The ability to accept, reject, or lock individual courses without discarding the entire schedule draft.
12. **Version Timeline & Undo:** Preserves all interactions and constraint changes, allowing full reversibility.